Supplementary Materials: Going Beyond Consistency: Target-oriented Multi-view Graph Neural Network

A Proof of Theorem 1

Proof. Theorem 1 presents the expected risk is bounded by empirical risk. We proof the theorem referring to Jensen's inequality, Markov inequality and Hoeffding's inequality. For $\mathcal{L}_{ce}$, we first use the variational inference process to change the measure of distribution, and then, we use Markov's inequality to calculate the bound of risk. In specific, defining $\Delta(\mathcal{L}_{ce}) = \widehat{\mathcal{L}}_{ce} - \mathcal{L}_{ce}$, we apply the variational inference trick and use Jensen's inequality to get the following inequality

$$
\begin{aligned}
&\Delta(\mathcal{L}_{ce}) \\
=&\widehat{\mathcal{L}}_{ce} - \mathcal{L}_{ce} \\
\leq&\mathbb{E}_{\mathcal{P}} \operatorname{KL}\Big(\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C} \mid \mathcal{G}(\mathbf{x}))\|\pi_c\Big) - \\
&2(n_l-1)\mathbb{E}_{\mathcal{J}}\mathbb{E}_{\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}|\mathcal{G}(\mathbf{x}))}\ln\frac{\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C} \mid \mathcal{G}(\mathbf{x}))}{\pi_c} + \\
&\ln\mathbb{E}_{\mathbf{c}\sim\pi_c} exp\Big(\mathbb{E}_{\mathcal{P}}\mathbf{y}\ln\big(\sigma\big(f^{\mathbf{\Theta}}(\mathbf{c})\big)\big)\Big) - \\
&\ln\mathbb{E}_{\mathbf{c}\sim\pi_c} exp\Big(\mathbb{E}_{\mathcal{J}}\mathbf{y}\ln\big(\sigma\big(f^{\mathbf{\Theta}}(\mathbf{c})\big)\big)\Big) \\
=&\mathbb{E}_{\mathcal{P}} \operatorname{KL}\Big(\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C} \mid \mathcal{G}(\mathbf{x}))\|\pi_c\Big) - \\
&\mathbb{E}_{\mathcal{J}}\mathbb{E}_{\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}|\mathcal{G}(\mathbf{x}))}\ln\frac{\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C} \mid \mathcal{G}(\mathbf{x}))}{\pi_c} + \\
&\ln\mathbb{E}_{\mathbf{c}\sim\pi_c} exp\Big(\Delta(E)\Big) \\
=&\mathbb{E}_{\mathcal{P}} \operatorname{KL}\Big(\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C} \mid \mathcal{G}(\mathbf{x}))\|\pi_c\Big) - \\
&\mathop{\mathbb{E}}_{\mathcal{J}} \operatorname{KL}\Big(\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C} \mid \mathcal{G}(\mathbf{x}))\|\pi_c\Big) + \\
&\ln\mathbb{E}_{\mathbf{c}\sim\pi_c} exp\Big(\Delta(E)\Big),
\end{aligned}
\tag{1}
$$

where the term $\Delta(E) = |\mathbb{E}_{\mathcal{P}}\mathbf{y}\ln\big(\sigma\big(f^{\mathbf{\Theta}}(\mathbf{c})\big)\big) - \mathbb{E}_{\mathcal{J}}\mathbf{y}\ln\big(\sigma\big(f^{\mathbf{\Theta}}(\mathbf{c})\big)\big)|$. Recall that Hoeffding's inequality and denoting P as the probability, we gain the following inequality

$$
P[\Delta(E) \geq \eta] \leq \exp(-2n_l)\eta^2. \tag{2}
$$

Then, denoting the density function of $\Delta(E)$ as $f\big(\Delta(E)\big)$, we have,

$$
\begin{aligned}
&P[\Delta(E) \geq \eta] = e^{-2n_l\eta^2} \\
\Rightarrow &\int_{\eta}^{\infty} f(\Delta(E))d\Delta(E) = e^{-2n_l\eta^2} \\
\Rightarrow &f(\eta) = 4n_l\eta e^{-2n_l\eta^2}.
\end{aligned}
\tag{3}
$$

Thus, we can gain

$$
\begin{aligned}
&\mathbb{E}_{\mathcal{P}}\left[\exp\left(2(n_l-1)\Delta^2(E)\right)\right] \\
=&\int_0^1 f(\Delta(E))\exp\left(2(n_l-1)\Delta^2(E)\right)d\Delta(E) \\
\leq&\int_0^1 4n_l\Delta(E)\exp\left(-2n_l\Delta^2(E)\right)\exp\left(2(n_l-1)\Delta^2(E)\right)d\Delta(E) \\
=&-n_l\exp\left(-2\Delta^2(E)\right)\Big|_0^1 \\
=&\left(1-e^{-2}\right)n_l < n_l.
\end{aligned}
\tag{4}
$$

Combining Eq. (1) and Eq. (4) and supposing $g(\Delta(E)) = \ln\mathbb{E}_{\pi_c}\left[\exp\left(2(n_l-1)\Delta^2(E)\right)\right]$, since we have

$$
\mathbb{E}_{\mathcal{P}}\mathbb{E}_{\pi_c}\left[\exp\left(2(n_l-1)\Delta^2(E)\right)\right] \leq n_l, \tag{5}
$$

through Markov's inequality, we further get,

$$
P_{(\mathcal{P})}[g(\Delta(E)) \geq \eta] \leq \frac{n_l}{e^{\eta}}. \tag{6}
$$

Suppose that $\eta = \ln(n_l/\epsilon)$ and with the probability of at least $1-\epsilon$, we have that for all π_c,

$$
\Rightarrow 2(n_l-1)\ln\mathbb{E}_{\pi_c}\left[\exp\left(\Delta^2(E)\right)\right] \leq \eta = \ln(n_l/\epsilon). \tag{7}
$$

Because $\Delta(E) \in [0,1]$, we have,

$$
\exp\left(\Delta^2(E)\right) \leq \Big(\exp\left(\Delta(E)\right)\Big)^2 \leq e\cdot\exp\left(\Delta^2(E)\right). \tag{8}
$$

The last inequality due to $\exp\left(\Delta^2(E)\right) - \Big(\exp\left(\Delta(E)\right)\Big)^2$ is a monotonically decreasing when $\Delta(E) \in [0,1]$. Then, according to Jensen's inequality, we conclude that

$$
\begin{aligned}
\ln\Big(\mathbb{E}_{\mathbf{c}\sim\pi_c}\exp\left(\Delta(E)\right)\Big)^2 &\leq \ln\mathbb{E}_{\mathbf{c}\sim\pi_c}\big(\exp\Delta(E)\big)^2 \\
&\leq \ln\Big(e\cdot\mathbb{E}_{\mathbf{c}\sim\pi_c}\exp\Delta^2(E)\Big) \\
&\leq \frac{1}{2(n_l-1)}\ln\frac{n_l}{\epsilon} + 1
\end{aligned}
$$

$$\Rightarrow \ln\Big(\mathbb{E}_{\mathbf{c}\sim\pi_c}\exp\big(\Delta(E)\big)\Big) \leq \frac{1}{4(n_l-1)}\ln\frac{n_l}{\epsilon}+\frac{1}{2}. \quad (9)$$

Then,

$$\begin{aligned}
&|\widehat{\mathcal{L}}_{ce}-\mathcal{L}_{ce}| \\
&\leq |\mathbb{E}_{\mathcal{P}}\,\mathrm{KL}\left(\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))\|\pi_c\right)- \\
&\mathbb{E}_{\mathcal{J}}\,\mathrm{KL}\left(\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))\|\pi_c\right)+\frac{1}{4(n_l-1)}\ln\frac{n_l}{\epsilon}+\frac{1}{2}| \\
&\leq |\mathbb{E}_{\mathcal{P}}\,\mathrm{KL}\left(\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))\|\pi_c\right)+ \\
&\mathbb{E}_{\mathcal{J}}\,\mathrm{KL}\left(\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))\|\pi_c\right)+\frac{1}{4(n_l-1)}\ln\frac{n_l}{\epsilon}+\frac{1}{2}| \\
&\leq \mathbb{E}_{\mathcal{P}}\,\mathrm{KL}\left(\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))\|\pi_c\right)+\frac{1}{4(n_l-1)}\ln\frac{n_l}{\epsilon}+B.
\end{aligned} \quad (10)$$

We prove the conclusion (9) in Theorem 1.

Similarly, for the prediction separation term, we define $\Delta(\mathcal{L}_{spe}^v) = \widehat{\mathcal{L}}_{spe}^v - \mathcal{L}_{spe}^v$, in which $\widehat{\mathcal{L}}_{spe}^v = \mathbb{E}_{\mathcal{P}}\mathbb{E}_{\mathbf{c}\sim\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}|\mathcal{G}(\mathbf{x}))}\mathbb{E}_{\mathbf{h}\sim\mathcal{D}_{tr}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}|G^v(\mathbf{x}))}\mathrm{I}\left[f^{\mathbf{\Theta}}(\mathbf{c})=f^{\mathbf{\Theta}}(\mathbf{h})\right]$. Different $\mathcal{L}_{ce}$, the prediction separation loss has an extra expectation on $\mathbf{h}$. We apply Jensen's inequality again and then use the variational inference trick to get the derivation results,

$$4(n_l-1)^2\Delta(\mathcal{L}_{spe}^v)^2 = 4(n_l-1)^2\Big(\widehat{\mathcal{L}}_{spe}^v-\mathcal{L}_{spe}^v\Big)^2. \quad (11)$$

Then, we discuss the term $\mathcal{L}_{spe}^v$ and $\widehat{\mathcal{L}}_{spe}^v$, respectively.

$$\begin{aligned}
\mathcal{L}_{spe}^v =&\mathop{\mathbb{E}}_{\mathcal{J}}\mathbb{E}_{\mathbf{c}\sim\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}|\mathcal{G}(\mathbf{x}))}\mathbb{E}_{\mathbf{h}\sim\mathcal{D}_{te}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}|G^v(\mathbf{x}))} \\
&\mathrm{I}\left[f^{\mathbf{\Theta}}(\mathbf{c})=f^{\mathbf{\Theta}}(\mathbf{h})\right] \\
=&\mathop{\mathbb{E}}_{\mathcal{J}}\Big[\mathbb{E}_{\mathbf{c}}\mathbb{E}_{\mathbf{h}}\ln\frac{\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))}{\pi_c}+ \\
&\mathbb{E}_{\mathbf{c}}\mathbb{E}_{\mathbf{h}}\ln\frac{\mathcal{D}_{te}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}\mid G^v(\mathbf{x}))}{\pi_{s_v}}+ \\
&\mathbb{E}_{\mathbf{c}}\mathbb{E}_{\mathbf{h}}\ln\frac{\pi_c}{\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))}\frac{\pi_{s_v}}{\mathcal{D}_{te}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}\mid G^v(\mathbf{x}))} \\
&\exp\Big(\mathrm{I}\left[f^{\mathbf{\Theta}}(\mathbf{c})=f^{\mathbf{\Theta}}(\mathbf{h})\right]\Big)\Big] \\
\leq&\mathop{\mathbb{E}}_{(\mathbf{x},\mathbf{y})\sim\mathcal{J}}\Big[\mathrm{KL}(\mathcal{D}_{te}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))||\pi_c)+ \\
&\mathrm{KL}(\mathcal{D}_{te}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}\mid G^v(\mathbf{x}))||\pi_{s_v})\Big]+ \\
&\ln\mathbb{E}_{\mathbf{c}\sim\pi_c}\mathbb{E}_{\mathbf{h}\sim\pi_{s_v}}\exp\Big(\mathbb{E}_{\mathcal{J}}\mathrm{I}\left[f^{\mathbf{\Theta}}(\mathbf{c})=f^{\mathbf{\Theta}}(\mathbf{h})\right]\Big).
\end{aligned} \quad (12)$$

Similarly,

$$\begin{aligned}
\widehat{\mathcal{L}}_{spe}^v =&\mathbb{E}_{\mathcal{P}}\mathbb{E}_{\mathbf{c}\sim\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}|\mathcal{G}(\mathbf{x}))}\mathbb{E}_{\mathbf{h}\sim\mathcal{D}_{tr}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}|G^v(\mathbf{x}))} \\
&\mathrm{I}\left[f^{\mathbf{\Theta}}(\mathbf{c})=f^{\mathbf{\Theta}}(\mathbf{h})\right] \\
=&\mathbb{E}_{\mathcal{P}}\Big[\mathbb{E}_{\mathbf{c}}\mathbb{E}_{\mathbf{h}}\ln\frac{\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))}{\pi_c}+ \\
&\mathbb{E}_{\mathbf{c}}\mathbb{E}_{\mathbf{h}}\ln\frac{\mathcal{D}_{tr}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}\mid G^v(\mathbf{x}))}{\pi_{s_v}}+ \\
&\mathbb{E}_{\mathbf{c}}\mathbb{E}_{\mathbf{h}}\ln\frac{\pi_c}{\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))}\frac{\pi_{s_v}}{\mathcal{D}_{tr}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}\mid G^v(\mathbf{x}))} \\
&\exp\Big(\mathrm{I}\left[f^{\mathbf{\Theta}}(\mathbf{c})=f^{\mathbf{\Theta}}(\mathbf{h})\right]\Big)\Big] \\
\leq&\mathop{\mathbb{E}}_{(\mathbf{x},\mathbf{y})\sim\mathcal{P}}\Big[\mathrm{KL}(\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))||\pi_c)+ \\
&\mathrm{KL}(\mathcal{D}_{tr}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}\mid G^v(\mathbf{x}))||\pi_{s_v})\Big]+ \\
&\ln\mathbb{E}_{\mathbf{c}\sim\pi_c}\mathbb{E}_{\mathbf{h}\sim\pi_{s_v}}\exp\Big(\mathbb{E}_{\mathcal{P}}\mathrm{I}\left[f^{\mathbf{\Theta}}(\mathbf{c})=f^{\mathbf{\Theta}}(\mathbf{h})\right]\Big).
\end{aligned} \quad (13)$$

Combining Eqs. (12), (13) into Eq. (11), we have,

$$\begin{aligned}
&4(n_l-1)^2\Delta(\mathcal{L}_{spe}^v)^2 = 4(n_l-1)^2\Big(\widehat{\mathcal{L}}_{spe}^v-\mathcal{L}_{spe}^v\Big)^2 \\
\leq&\Big(2(n_l-1)\big(\mathbb{E}_{\mathcal{P}}\,\mathrm{KL}(\mathcal{D}_{tr}^{\mathbf{\Phi}^c,\mathbf{\Omega}}(\mathbf{C}\mid\mathcal{G}(\mathbf{x}))||\pi_c)+ \\
&\mathbb{E}_{\mathcal{P}}\,\mathrm{KL}(\mathcal{D}_{tr}^{\mathbf{\Phi}^{s_v}}(\mathbf{H}^{s_v}\mid G^v(\mathbf{x}))||\pi_{s_v})+ \\
&\ln\mathbb{E}_{\mathbf{c}\sim\pi_c}\mathbb{E}_{\mathbf{h}\sim\pi_{s_v}}\exp(2(n_l-1)\Delta(M))\Big)\Big)^2.
\end{aligned} \quad (14)$$

Here, $\Delta(M) = |\mathbb{E}_{\mathcal{P}}\mathrm{I}\left[f^{\mathbf{\Theta}}(\mathbf{c})=f^{\mathbf{\Theta}}(\mathbf{h})\right] - \mathbb{E}_{\mathcal{J}}\mathrm{I}\left[f^{\mathbf{\Theta}}(\mathbf{c})=f^{\mathbf{\Theta}}(\mathbf{h})\right]|$. The rest of the proof is similar to the proof of $\mathcal{L}_{ce}$, thus we get the theoretical results of Eq. (10) in Theorem 1. □

B Training Algorithm

Algorithm 1 shows the algorithm flow of TGNN in the main paper.

C More Experiments

C.1 Datasets

his subsection presents the datasets utilized in the main paper, which can be categorized into three types: multi-relational graphs, multi-attribute and multi-modal data. The statistical descriptions of these datasets are shown in Tables 1 to 2. Detailed explanations of each dataset are provided below.

- **Multi-relational graphs**
  - **ACM** is a citation network expressing the relationship between papers, which contains three types of nodes and two types of edges.
  - **DBLP** a bibliography website used in the field of computer science, including four types of nodes and three types of edges each which is the link between authors.

Algorithm 1 TGNN

Input: Node feature $\mathbf{X}$, Multi-view adjacency matrices $\{\mathbf{A}^v\}_{v=1}^V$, the number of layer L, hyperparameters α and θ.
Parameter: Learnable parameters $\mathbf{\Phi}^c, \mathbf{\Omega}, \{\mathbf{\Phi}^{s_v}\}_{v=1}^V, \mathbf{\Theta}$.
Output: Representation $\mathbf{C}$.

while not convergent **do**
  // Forward Propagation.
  Obtain the determinative representation $\mathbf{C}$ with Eqs. (2), (3).
  Input $\mathbf{C}$ into the classifier $f^{\mathbf{\Theta}}$.
  for $v = 1 \to V$ **do**
    Calculate the incidental representation $\mathbf{H}^{s_v}$ for view v with Eq. (2).
    Input $\mathbf{H}^{s_v}$ into the classifier $f^{\mathbf{\Theta}}$.
  end for
  // Backward Propagation.
  for $v = 1 \to V$ **do**
    Update $\{\mathbf{\Phi}^c, \mathbf{\Omega}\}$ with losses $\mathcal{L}_{sha}^v$, $\mathcal{L}_{cdo}^v$, $\mathcal{L}_{KL}^v$.
    Update $\{\mathbf{\Phi}^{s_v}, \mathbf{\Theta}\}$ with $\mathcal{L}_{cdo}^v$ and $\mathcal{L}_{KL}^v$.
  end for
end while
return Representation $\mathbf{C}$.

  - **YELP** comes from merchant review website containing four types of nodes and three types of edges, where each edge linking two businesses.

- **Multi-attribute data**
  - **Animals** is a large dataset consisting of 10,158 images across 50 classes, with two types of deep features extracted using DECAF and VGG19.
  - **HW** contains handwritten digits in 10 classes, represented by six feature views: Profile Correlation, Fourier Coefficients, Karhunen-Loeve Coefficients, Morphological features, Intensity-Averaged features, and Zernike Moments.
  - **MNIST** is a handwritten digits dataset with 10,000 samples, where each image is affected by white Gaussian noise of varying intensity.
- **Multi-modal data**
  - **BDGP** is a dataset of 2,500 images of Drosophila embryos categorized into 5 classes, with each image represented by 1,750-dimensional visual features and 79-dimensional textual features.
  - **esp-game** contains 20,770 images, each annotated with multiple descriptions by players. We select 13,855 images described with more than four tags, categorized into a total of 7 classes.
  - **Flickr** consists of 25,000 images with 1,386 text tags collected from Flickr. For experiments, 12,154 images covering 7 categories are selected, including two modalities: image and text.

C.2 Methods

- **Methods for multi-attribute and multi-modal classification**

Table 1: Statistics of multi-relational graphs.

Datasets	Nodes	Features	Views	Classes
ACM	3,025	1,870	2	3
DBLP	4,057	334	3	4
YELP	2,614	82	3	3

Table 2: Statistics of multi-attribute and multi-modal datasets.

Datasets	Nodes	Features	Views	Classes
Animals	10,158	8,192	2	50
HW	2,000	1,715	6	10
MNIST	10,000	48	3	10
BDGP	2,500	1,829	2	5
esp-game	11,032	200	2	7
Flickr	12,154	200	2	7

  - **HAN** leverages attention at both the node level and semantic level on multiplex networks to capture the significance of nodes and meta-paths, generating node representations in a hierarchical way.
  - **DMGI** is an unsupervised attributed multiplex network that integrates node embeddings from multiple relations to learn high-quality representations. This is achieved through a consensus regularization framework and a universal discriminator for downstream tasks.
  - **IGNN** is a graph learning framework that uses a fixed-point equilibrium equation and Perron-Frobenius theory to iteratively perform graph convolutional aggregation until convergence, specifically for node classification tasks.
  - **MRGCN** is a tensor-constrained heterogeneous graph-based convolutional learning method, which considers the correlation between different relations.
  - **SSDCM** aims to maximize the mutual information between local and contextualized global graph summaries. It employs cross-layer links to impose regularization on the node embeddings.
  - **MHGCN** automatically learns useful relation-aware topological structural signals through multiplex relation aggregation and multi-layer graph convolution for graph representation learning tasks.
  - **AMOGCN** discovers meta-paths that contain multi-hop neighbors and fuses adjacency matrices of different orders to construct a comprehensive multi-order adjacency matrix.
  - **ECMGD** constructs a graph diffusion model with energy constraint to devise a feature propagation formula considering inter- and intra-view feature flow.
- **Methods for multi-relational classification**
  - **Co-GCN** introduces GCN into multi-view learning and obtains multi-view spectral information by adaptively combining Laplacian matrices.

- **HLR-M^{2}VS** constructs a unified tensor space to jointly explore the relationships among multiple views, leveraging a local geometric structure.
- **ERL-MVSC** integrates diversity, sparsity, and consensus to effectively handle multi-view data with limited labels.
- **DSRL** uses a deep sparse regularizer learning model to adaptively learn data-driven sparse regularizers for multi-view clustering and semi-supervised classification.
- **PDMF** learns the relations and the auxiliary representation via pre-training to tune the mappings from the original data to comprehensive representations.
- **IMvGCN** introduces multi-view reconstruction errors paired with Laplace embeddings to capture both independence and consistency across views.
- **GEGCN** aims to extract the consistency and specificity among topology structures generated by multi-view attribute and leverages a threshold function to filter noise in graphs.

C.3 More Experimental Results

Robustness Analysis. In the robustness analysis, we employ the state-of-the-art adversarial attack method, Mettack [Zugner and Gunnemann, 2019], to introduce perturbations into the graph structure. Mettack specifically targets the graph topology during training, focusing on the largest connected components for the attack. The perturbation rate refers to the proportion of edges that are altered in the graph; for example, a 25% perturbation rate means that 25% of the original edges are replaced with adversarial edges. The results of applying Mettack to DBLP are presented in Figure 1.

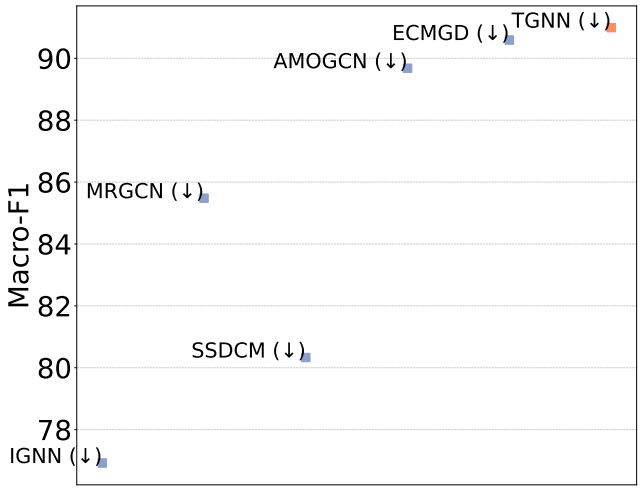


Figure 1: Performance of various GNN-based methods after the attack on DBLP.

Visualization Results. We provide T-SNE visualizations of other GNN-based competitors on the HW dataset, as shown in Figure 2. Compared to these visualizations, TGNN demonstrates clearer classification results, where samples of the same class are clustered together, and samples of different classes are effectively separated.

Parameter Sensitivity. Figure 3 presents the results of TGNN with varying values of α and θ on the DBLP, HW, esp-game, and Flickr datasets. Regarding α, it is evident that performance is significantly influenced by its value, and the trend of change varies across different datasets. For θ, the optimal performance for the HW and esp-game datasets is achieved at $\theta = 0.1$. This could be attributed to the substantial disparity between semantics in these two datasets, which causes the model to struggle with adjusting its network parameters effectively during training, ultimately leading to suboptimal learning strategies.

Ablation Study. Figure 4 presents the performance of TGNN and its variants on the DBLP, YELP, esp-game, Flickr, and Animals datasets. It is clear that BaseGNN performs the worst, while TGNN achieves the best results compared to the other variants, demonstrating the effectiveness of the proposed TGNN model.

References

[Zugner and Gunnemann, 2019] Daniel Zugner and Stephan Gunnemann. Adversarial attacks on graph neural networks via meta learning. In *Proceedings of the Tnternational Conference on Learning Representations*, 2019.

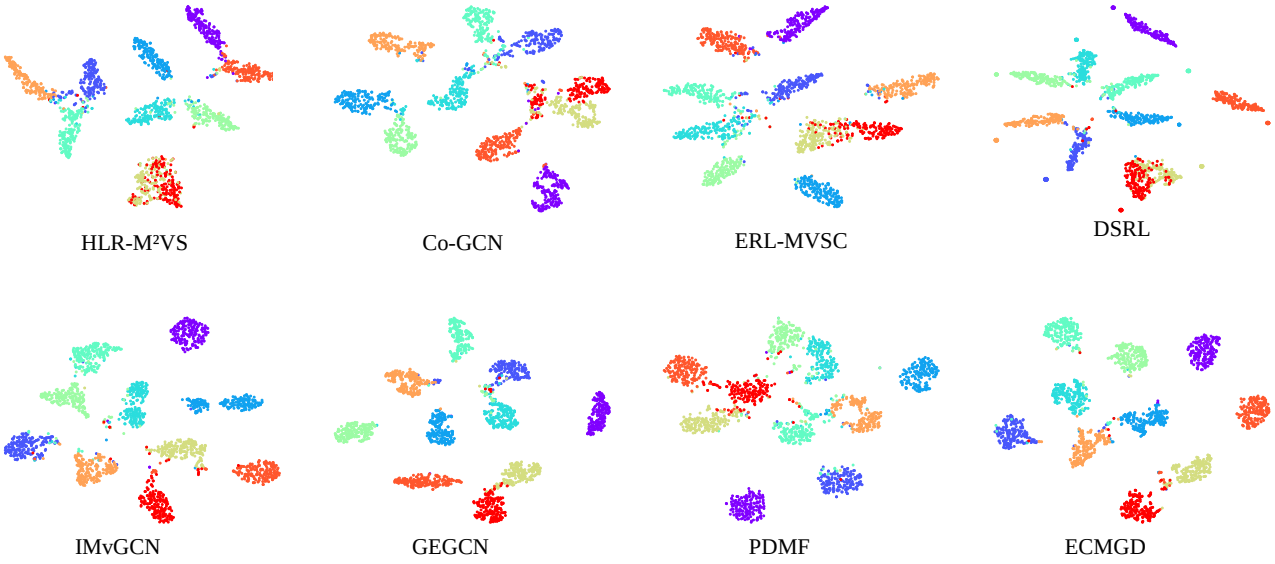


Figure 2: Visualizations of competitors on HW.

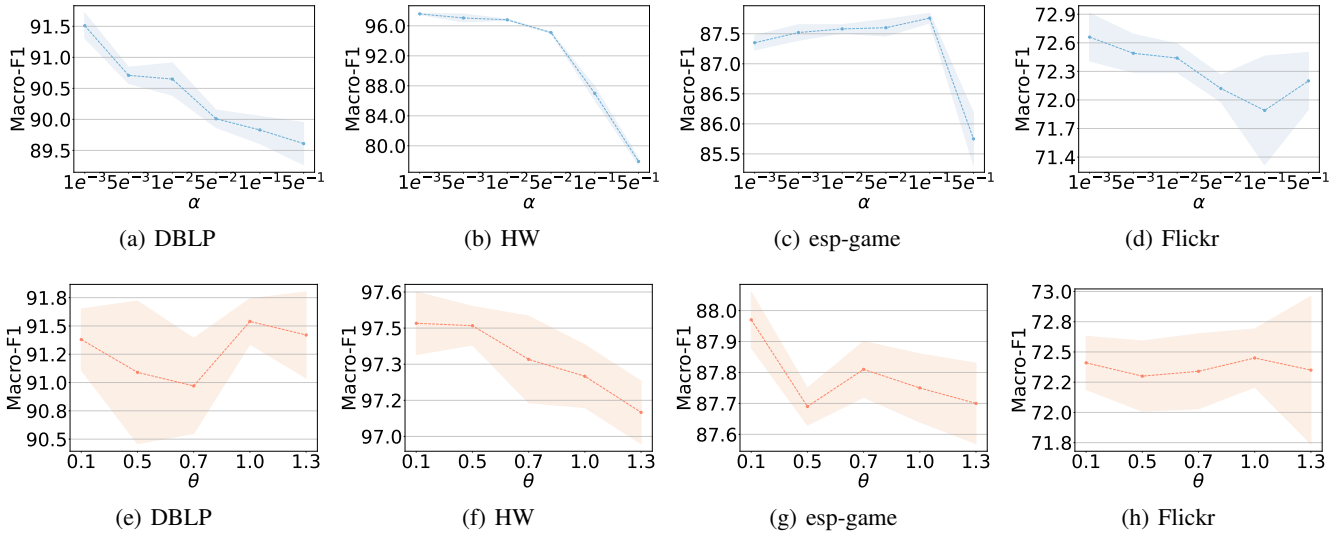


Figure 3: Parameter sensitivity on four multi-view datasets, where (a)-(d) show the performance of TGNN w.r.t. α and (e)-(h) display the performance of TGNN w.r.t. θ.

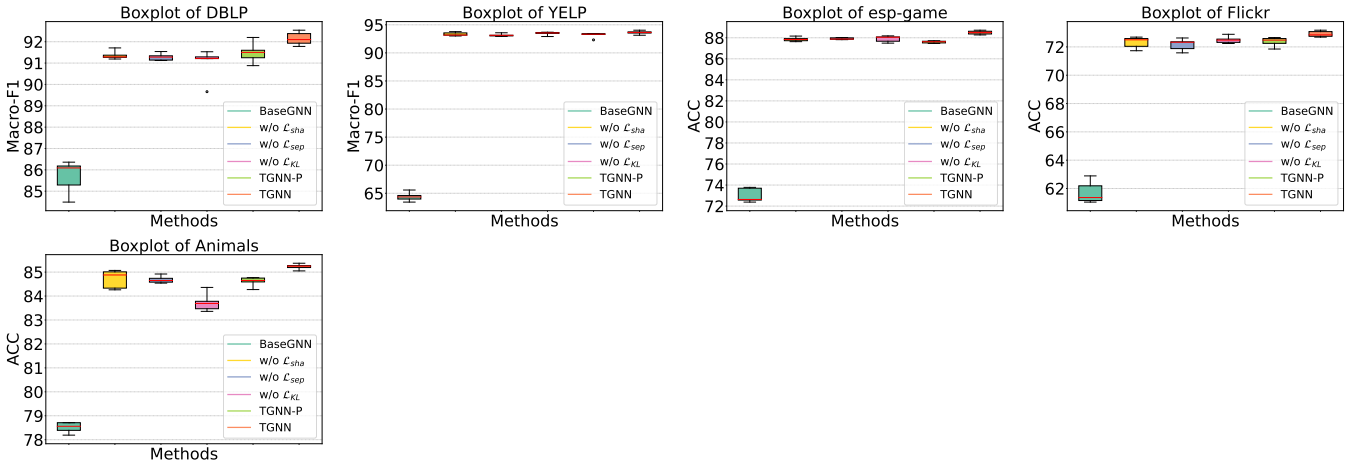


Figure 4: Results of TGNN and its variants on remaining datasets.